

1. Administrative & Study Identification

Full Study Title: Pembrolizumab/Placebo Plus Trastuzumab Plus Chemotherapy in Human Epidermal Growth Factor Receptor 2 Positive (HER2+) Advanced Gastric or Gastroesophageal Junction (GEJ) Adenocarcinoma (MK-3475-811/KEYNOTE-811) **Short Title/Acronym:** KEYNOTE-811 **Protocol Number:** MK-3475-811 **NCT Number:** NCT03615326 **Posting ID:** 926101324492067992 **Sponsor Name:** Merck **Investigational Product:** Pembrolizumab (Keytruda) **Phase of Development:** Phase III **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate if pembrolizumab plus trastuzumab in combination with standard of care (SOC) chemotherapy is superior to placebo plus trastuzumab and SOC chemotherapy in terms of Progression-Free Survival (PFS) and Overall Survival (OS). **Secondary Objectives:** To evaluate the Objective Response Rate (ORR), Duration of Response (DOR), and the safety and tolerability (Number of Participants Who Experienced an Adverse Event).

2.2 Background & Rationale Treatment options for patients with HER2-positive advanced gastric cancer are limited, and the prognosis is generally poor. Combination therapy with the anti-HER2 antibody trastuzumab plus fluoropyrimidine and platinum is the current standard of care. This study is conducted to address whether combining an anti-PD-1 therapy (pembrolizumab) with anti-HER2 therapy and chemotherapy will result in augmented T-cell activation and antibody-dependent cell-mediated cytotoxicity, thereby significantly improving clinical outcomes such as overall survival for this patient population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1

3.2 Planned Enrollment & Duration Target \$N\$: 738 participants **Number of Sites:** Global multicenter **Estimated Study Start:** 2018 **Estimated Primary Completion:** Q4-2024 (Interim and Final Analyses have met primary endpoints)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically or cytologically confirmed diagnosis of previously untreated, locally advanced unresectable or metastatic human epidermal growth factor receptor 2 (HER2) positive gastric or gastroesophageal junction (GEJ) adenocarcinoma. HER2+ must be centrally confirmed (IHC 3+ or IHC 2+/ISH or FISH positive). 3. Life expectancy > 6 months with measurable disease per RECIST v1.1. 4. Adequate organ function and an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1.

4.2 Exclusion Criteria 1. Previously treated with systemic therapy for metastatic disease. 2. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 3. Active autoimmune disease that has required systemic treatment in the past 2 years or a documented history of severe hypersensitivity.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Pembrolizumab** (200 mg) or **Placebo** administered intravenously (IV) on Day 1 of each 3-week cycle. * **Trastuzumab** (8 mg/kg loading dose, followed by 6 mg/kg maintenance dose) administered IV on Day 1 of each 3-week cycle. * **SOC Chemotherapy (Investigator's Choice):** * **FP:** 80 mg/m² cisplatin (ATC code: L01XA01, Trade Name: Platinol) IV on Day 1 + 800 mg/m² 5-fluorouracil (5-FU) IV on Days 1-5 of each 3-week cycle; OR * **CAPOX:** 1000 mg/m² capecitabine orally BID on Days 1-14 + 130 mg/m² oxaliplatin IV on Day 1 of each 3-week cycle; OR * **SOX (Japan Cohort):** S-1 (tegafur, 5-chloro-2,4-dihydroxypyridine [CDHP], and potassium oxonate [Oxo]) administered orally BID according to Body Surface Area (BSA) on Days 1-14 + 130 mg/m² oxaliplatin administered IV on Day 1 of each 3-week cycle. **Route of Administration:** Intravenous (IV) and Oral. **Duration of Treatment:** Up to 2 years, or until intolerable toxicity or disease progression.

5.2 Concomitant & Prohibited Therapy Permitted: Standard pre-medications for chemotherapy and prophylactic treatments for infusion reactions. **Prohibited:** Concurrent administration of other investigational anti-cancer agents, systemic immunosuppressive therapy (outside of standard pre-medications), or live vaccines.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor biopsy for PD-L1/HER2 testing, Baseline Scans
Baseline	Day 0	Randomization, First Dose of

pembrolizumab/placebo + trastuzumab + chemotherapy | | **Interim** | Every 3 Weeks | Dosing (Q3W), Safety Labs, Efficacy Assessment (Tumor imaging per protocol) | | **End of Study** | Up to 2 Years | Final Efficacy Scans, Safety follow-up (30/90 days), Survival follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint 1. **Progression-Free Survival (PFS)**: Per RECIST 1.1 Assessed by Blinded Independent Central Review (BICR). 2. **Overall Survival (OS)**: Time from randomization to death due to any cause.

7.2 Secondary & Exploratory Endpoints 1. Objective Response Rate (ORR) Per RECIST 1.1 Assessed by BICR. 2. Duration of Response (DOR) Per RECIST 1.1 Assessed by BICR. 3. Number of Participants Who Experienced an Adverse Event (AE) (Evaluating safety and tolerability).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Graded per CTCAE v4.0 and continuously monitored throughout the trial and for 30 or 90 days following the cessation of treatment. **Serious Adverse Events (SAEs)**: Reported within standard 24-hour safety timelines. **Data Monitoring Committee (DMC)**: External independent committee to evaluate interim safety and efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intention-to-Treat (ITT) population for all primary efficacy measures (Global Cohort). Participants in the Japan-specific SOX Cohort were not included in the primary efficacy analysis per the SAP. **Sample Size Justification**: Powered to detect a statistically significant and clinically meaningful improvement in OS and PFS against the placebo control arm. **Handling of Missing Data**: Standard oncology data handling, censoring rules applied for PFS and OS endpoints as per RECIST 1.1 guidelines.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan**: Standard periodic onsite and centralized remote monitoring visits to ensure data fidelity. **Audit Trail**: Continuous tracking via eCRF locking and blinded independent central imaging reviews.

11. Study Identifiers & Governance

Primary ID: {{MK-3475-811}} **Registry Number:** {{NCT03615326}}
Posting ID: {{926101324492067992}} **Sponsor/Lead Organization:** {{Merck}} **Study Title:** {{Pembrolizumab/Placebo Plus Trastuzumab Plus Chemotherapy in Human Epidermal Growth Factor Receptor 2 Positive (HER2+) Advanced Gastric or Gastroesophageal Junction (GEJ) Adenocarcinoma (MK-3475-811/KEYNOTE-811)}} **Official Regulatory Title:** {{A Phase III, Randomized, Double-blind Trial Comparing Trastuzumab Plus Chemotherapy and Pembrolizumab With Trastuzumab Plus Chemotherapy and Placebo as First-line Treatment in Participants With HER2 Positive Advanced Gastric or Gastroesophageal Junction Adenocarcinoma (KEYNOTE 811)}}}

12. Clinical Framework (Gastric Cancer Specific)

Primary Condition: {{HER2-positive Gastric Neoplasms and Gastroesophageal Junction Adenocarcinoma (Semantic Type: Neoplastic Process)}} **Diagnosis Threshold:** {{HER2-positive (IHC 3+ or IHC 2+/ISH > 2.0) and PD-L1 expression evaluation (CPS \$ \ge\$ 1)}} **Medical Methodology:** {{Anti-PD-1 + Anti-HER2 + Fluoropyrimidine/Platinum Chemotherapy}} **Optimization Target:** {{Overall Survival (OS) and Progression-Free Survival (PFS) extension}}}

13. Study Procedures & Methodology

Management Pathway: {{Standard Oncology Pathway for mG/GEJc}} **Education Protocol (HCP):** {{Protocol-specific dosing, immune-mediated AE management, and RECIST criteria training}} **Education Protocol (Patient):** {{Informed consent, infusion schedule education, and side-effect reporting parameters}} **Quality Audit Mechanism:** {{Blinded Independent Central Review (BICR) for primary imaging endpoints}}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to 2 years of active treatment + Long-term Survival Follow-up}} **Onsite Visit Cadence:** {{Every 3 Weeks (Q3W) for dosing}} **Remote Monitoring Frequency:** {{As clinically indicated during survival follow-up}} **Checklist Review Interval:** {{Prior to each 3-week dosing cycle}}}

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator				[Name]		____		[DD-MMM-YYYY]								

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |